ECE 647: Lecture 24

Second Approach: Decrease of Norm

Lemma 1. *If f is convex, then*

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq 0 \tag{1}$$

Note: This holds even if $\nabla f(\cdot)$ is a subgradient. A mapping of $\nabla f(\cdot)$ that satisfies (1) is called a *monotone mapping*.

For 1-dim:

$$(f'(x) - f'(y))(x - y) \geq 0$$
$$\Leftrightarrow f'(x) \geq f'(y) \quad \text{when } x > y \quad \text{(monotonicity)}$$
$$\Leftrightarrow f''(x) \geq 0$$

Proof:

From the first-order condition of convexity:

$$f(y) \geq f(x) + \nabla f(x)(y - x)$$

$$f(x) \geq f(y) + \nabla f(y)(x - y)$$

Summing them:

$$(\nabla f(y) - \nabla f(x))^T (y - x) \geq 0 \quad \#$$

If the derivative is Lipschitz-continuous, then a stronger version can be shown.

Lemma 2. *Let f be a convex and differentiable function such that*

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2 \quad \forall x, y \in \mathbb{R}^n$$

Then,

$$(\nabla f(x) - \nabla f(y))^T(x - y) \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|_2^2 \quad \forall x, y \in \mathbb{R}^n \qquad (\star\star)$$

Note: If a mapping $\nabla f(\cdot)$ satisfies $(\star\star)$, the inverse mapping

$$\nabla f(x) \mapsto x$$

is called *strongly monotone*.

We will prove this lemma later.

Theorem 1. *Assume that*

$$(\nabla f(x) - \nabla f(y))^T(x - y) \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|_2^2 \quad \forall x, y \in \mathbb{R}^n$$

Further, assume that $0 < \eta < \frac{2}{L}$, and that there exists at least one point $x^\star$ such that

$$\nabla f(x^\star) = 0$$

Then the sequence $x(t)$ generated by the iterative process

$$x(t+1) = x(t) - \eta\nabla f(x(t))$$

converges, and the limit x_0 satisfies $\nabla f(x_0) = 0$.

Proof:

Let $x^\star$ be an optimal solution, i.e., $\nabla f(x^\star) = 0$.

$$\|x(t+1)-x^\star\|^2 = \|x(t)-x^\star\|^2+2(x(t+1)-x(t))^T(x(t)-x^\star)+\|x(t+1)-x(t)\|^2$$

Note that

$$(x(t+1) - x(t))^T(x(t) - x^\star) = -\eta(\nabla f(x(t)) - \nabla f(x^\star))(x(t) - x^\star)$$

$$\leq -\frac{\eta}{L}\|\nabla f(x(t))\|^2$$

Therefore,

$$\|x(t+1) - x^\star\|^2 \leq \|x(t) - x^\star\|^2 - \frac{2\eta}{L}\|\nabla f(x(t))\|^2 + \eta^2\|\nabla f(x(t))\|^2$$

$$= \|x(t) - x^\star\|^2 - \left(\frac{2\eta}{L} - \eta^2\right)\|\nabla f(x(t))\|^2$$

If $\eta < \frac{2}{L}$, then $\alpha = \frac{2\eta}{L} - \eta^2 > 0$

Summing over t:

$$\|x(t+1) - x^\star\|^2 \leq \|x(0) - x^\star\|^2 - \alpha \sum_{s=0}^{+\infty} \|\nabla f(x(s))\|^2$$

Since the left-hand side is non-negative, this implies

$$\sum_{s=0}^{+\infty} \|\nabla f(x(s))\|^2 < +\infty$$

$$\Rightarrow \nabla f(x(t)) \to 0$$

as $t \to \infty$.

$\square$

Q: Why does $x(t)$ converge?

A: (Key technique) Assume that a subsequence converges, i.e.,

$$x(t_h) \to x_0 \quad \text{as } h \to +\infty$$

where

$$t_h \to +\infty \quad \text{as } h \to +\infty$$

We must then have $\nabla f(x_0) = 0$ due to the continuity of $\nabla f(\cdot)$.

Now, replace $x^\star$ by x_0:

$$\lim_{t\to\infty} \|x(t) - x_0\|^2 \leq \lim_{h\to\infty} \|x(t_h) - x_0\|^2 = 0$$

$$\Rightarrow \quad x(t) \to x_0 \quad \text{as } t \to +\infty$$

Proof of Strong Monotonicity Lemma

Assume that ∇f is Lipschitz:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y$$

We want to show that:

$$(\nabla f(x) - \nabla f(y))^T(x - y) \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2$$

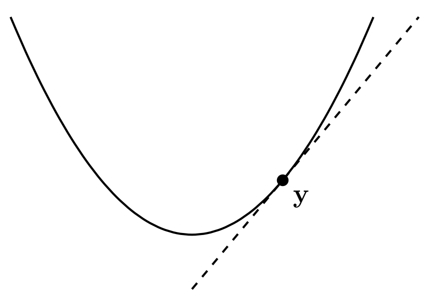
Note that by the growth lemma, for any convex function f with Lipschitz gradient:

$$f(x) \leq f(y) + \nabla f(y)(x - y) + \frac{L}{2}\|x - y\|^2 \qquad (\star)$$

for any convex function and $\nabla f(y)$ is Lipschitz.

We will show that:

$$f(x) \geq f(y) + \nabla f(y)^T(x - y) + \frac{1}{2L}\|\nabla f(x) - \nabla f(y)\|^2$$



Fix y.

Let

$$\tilde{f}(x) = f(x) - f(y) - \nabla f(y)^T(x - y)$$

We have

$$\tilde{f}(y) = 0, \quad \nabla\tilde{f}(x) = \nabla f(x) - \nabla f(y), \quad \nabla\tilde{f}(y) = 0$$

Hence y is the minimum of $\tilde{f}$.

Let:

$$z = x - \frac{1}{L}\nabla \tilde{f}(x)$$

Using $(\star)$ on $\tilde{f}(\cdot)$ have:

$$0 \leq \tilde{f}(z) \leq \tilde{f}(x) + \nabla \tilde{f}(x)\left(-\frac{1}{L}\nabla \tilde{f}(x)\right) + \frac{L}{2} \cdot \frac{1}{L^2}\left\|\nabla \tilde{f}(x)\right\|^2$$

Using the definition of $\tilde{f}(x)$, we have:

$$f(x) - f(y) - \nabla f(y)^T(x-y) - \frac{1}{2L}\|\nabla f(x) - \nabla f(y)\|^2 \geq 0$$

Interchanging the role of x and y, we can show that:

$$f(y) - f(x) - \nabla f(x)^T(y-x) - \frac{1}{2L}\|\nabla f(x) - \nabla f(y)\|^2 \geq 0$$

Combining these two inequalities, we have:

$$(\nabla f(y) - \nabla f(x))^T(y-x) \geq \frac{1}{L}\|\nabla f(y) - \nabla f(x)\|^2 \qquad \square$$

Geometric Convergence

- With smoothness, we can ensure convergence, but the speed of convergence may be slow.
- Adding **strong convexity** enables us to prove *geometric descent*.

$$\|\nabla f(x) - \nabla f(y)\|_2 \geq \alpha\|x-y\|_2$$

Intuitively, this ensures that when $x(t)$ is far from $x^\star$, the improvement of $\|x(t+1) - x^\star\|^2$ is directly related to $\|x(t) - x^\star\|^2$.

$$\|x(t+1) - x^\star\|^2 \leq \left(1 - \left(\frac{2\eta}{L} - \eta^2\right)\alpha\right)\|x(t) - x^\star\|^2 \qquad (\star)$$

The result above is stronger (i.e., faster descent).

Lemma 3. *Let f be an L-smooth and α-strongly convex function on $\mathbb{R}^n$. Then for all $x, y \in \mathbb{R}^n$,*

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq \frac{\alpha L}{\alpha + L} \|x - y\|^2 + \frac{1}{\alpha + L} \|\nabla f(x) - \nabla f(y)\|^2$$

Proof

Since f is α-strongly convex, we can define:

$$\varphi(x) = f(x) - \frac{\alpha}{2} \|x\|^2$$

is still convex. Further, we can show that $\varphi(x)$ is $(L - \alpha)$-smooth. Using the earlier lemma for smooth functions, we have:

$$[\nabla \varphi(x) - \nabla \varphi(y)]^T (x - y) \geq \frac{1}{L - \alpha} \|\nabla \varphi(x) - \nabla \varphi(y)\|^2$$

$$\Leftrightarrow [\nabla f(x) - \nabla f(y)]^T (x - y) - \alpha (x - y)^T (x - y) \geq \frac{1}{L - \alpha} \|\nabla f(x) - \nabla f(y) - \alpha (x - y)\|^2$$

$$= \frac{1}{L - \alpha} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{\alpha^2}{L - \alpha} \|x - y\|^2 - \frac{2\alpha}{L - \alpha} (\nabla f(x) - \nabla f(y))^T (x - y)$$

$$\Leftrightarrow \quad \frac{L + \alpha}{L - \alpha} (\nabla f(x) - \nabla f(y))^T (x - y) \geq \frac{1}{L - \alpha} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{L\alpha}{L - \alpha} \|x - y\|^2$$

The result of the Lemma then follows.

Theorem 2. *Let f be an L-smooth and α-strongly convex function on $\mathbb{R}^n$. Then, by setting a step size*

$$\gamma < \frac{2}{\alpha + L},$$

gradient descent satisfies

$$\|x(t) - x^\star\|^2 \leq \left(1 - \frac{2\gamma \alpha L}{\alpha + L}\right)^t \|x(0) - x^\star\|^2.$$

Proof

Starting with the norm approach again.

Let $x^\star$ be one optimal solution, i.e., $\nabla f(x^\star) = 0$.

$$\|x(t+1)-x^\star\|^2 = \|x(t)-x^\star\|^2+2(x(t+1)-x(t))^T(x(t)-x^\star)+\|x(t+1)-x(t)\|^2$$

Note that

$$(x(t+1) - x(t))^T(x(t) - x^\star) = -\gamma\left(\nabla f(x(t)) - \nabla f(x^\star)\right)^T(x(t) - x^\star)$$

$$\leq -\frac{\gamma\alpha L}{\alpha + L}\|x(t) - x^\star\|^2 - \frac{\gamma}{\alpha + L}\|\nabla f(x(t)) - \nabla f(x^\star)\|^2$$

Hence,

$$\|x(t+1)-x^\star\|^2 \leq \left(1 - \frac{2\gamma\alpha L}{\alpha + L}\right)\|x(t)-x^\star\|^2+\left(\gamma^2 - \frac{2\gamma}{\alpha + L}\right)\|\nabla f(x(t))-\nabla f(x^\star)\|^2$$

$$\leq \left(1 - \frac{2\gamma\alpha L}{\alpha + L}\right)\|x(t) - x^\star\|^2$$

note that $\left(\gamma^2 - \frac{2\gamma}{\alpha+L}\right) \leq 0$ if $\gamma < \frac{2}{\alpha+L}$.

The results of the Theorem then follows.

Note:

- For a faster convergence speed, we want
$$1 - \frac{2\gamma\alpha L}{\alpha + L}$$
to be smaller, or γ to be larger. set
$$\gamma = \frac{2}{\alpha + L}$$
- Compare with ($\star$) at $\gamma = \frac{2}{\alpha+L}$:
$$1 - \gamma\left(\frac{2}{L} - \gamma\right)\alpha = 1 - \frac{2\gamma\alpha^2}{L(\alpha + L)}$$

- When $\frac{\alpha}{L^2}$ is small, this quantity is much closer to 1.
- Using the inequality

  $$f(x) - f(x^\star) \leq \frac{L}{2}\|x - x^\star\|^2,$$

  we have

  $$f(x(t)) - f(x^\star) \leq \frac{L}{2}\left(1 - \frac{2\gamma\alpha L}{\alpha + L}\right)^t \|x(0) - x^\star\|^2.$$

- This is usually referred to as *linear convergence*.
- In contrast to *quadratic convergence* for second-order (e.g., Newton's) algorithms.